Structural Vibration Analyzer Project
Experimental Update Report – Forced Vibration Test at 20 Hz

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1 Introduction

This report presents an updated analysis of vibration data collected using the current Structural Vibration Analyzer setup. The system uses an ESP32 microcontroller, a PCA9548A I2C multiplexer, and three MPU6050 accelerometers. The objective of this test was to evaluate the vibration behavior of a 3D printer structure under a forced vibration condition of approximately:

$$f_{forcing} = 20 \text{ Hz}$$

The three sensors were mounted as follows:

- **Sensor 1 (s1):** mounted at the base/frame region.
- **Sensor 2 (s2):** mounted on the upper/opposite side of the frame.
- **Sensor 3 (s3):** mounted on the printer head/extruder assembly.

All sensors were mounted using rigid screw-mounted fixtures. This was done to reduce unwanted movement between the sensor and the structure and to improve the reliability of the measured vibration response.

2 Sampling Quality

The data was collected for approximately 92.629 seconds with 74,103 samples. The estimated sampling rate was:

$$f_s = 799.99 \text{ Hz}$$

The Nyquist frequency is therefore:

$$f_N = \frac{f_s}{2} = 399.99 \text{ Hz}$$

This means the FFT results are valid up to approximately 400 Hz.

Parameter	Value
Estimated sampling rate	799.99 Hz
Mean sampling interval	0.00125002 s
Minimum sampling interval	0.00121200 s
Maximum sampling interval	0.00249900 s
Sampling interval standard deviation	0.00000465 s
Duration	92.629 s
Number of samples	74103
Nyquist frequency	399.99 Hz
Estimated dropped samples	1
Repeated or reversed sample indices	0

Table 1: Sampling quality summary.

The sampling quality was very good. Only one dropped sample was estimated during the complete test. This indicates that the ESP32 binary logging method is working correctly at approximately 800 Hz.

3 Time-Domain Results

3.1 Sensor 1 – Base/Frame

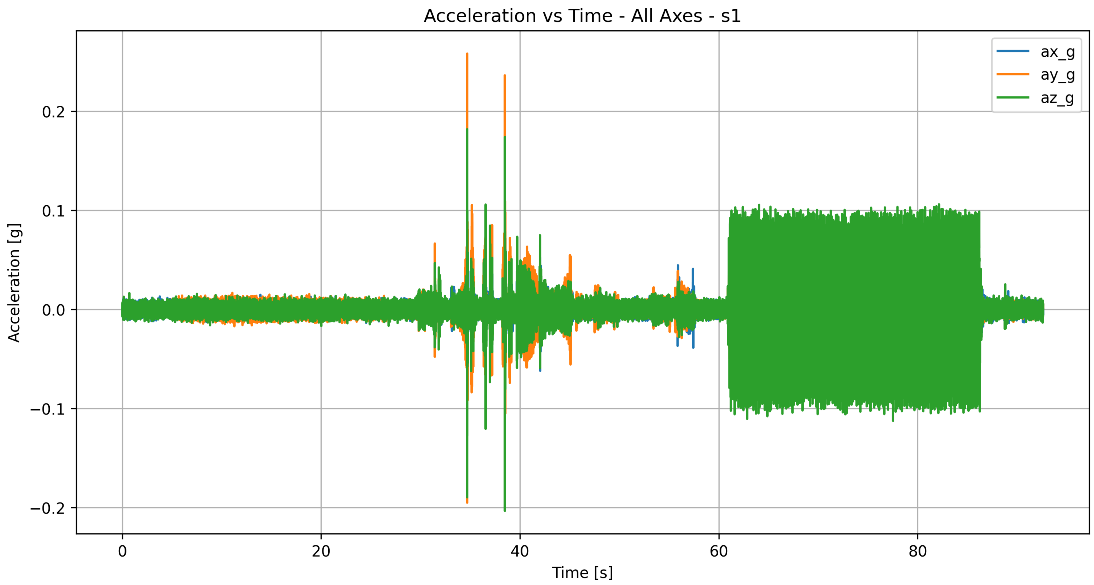

Figure 1: Acceleration vs time – all axes – Sensor 1.

Sensor 1 showed the lowest vibration levels of the three sensors. The measured accelerations stayed relatively small compared with the extruder sensor. This is expected because Sensor 1 was mounted near the base/frame, where the structure is generally more rigid.

3.2 Sensor 2 – Upper Frame

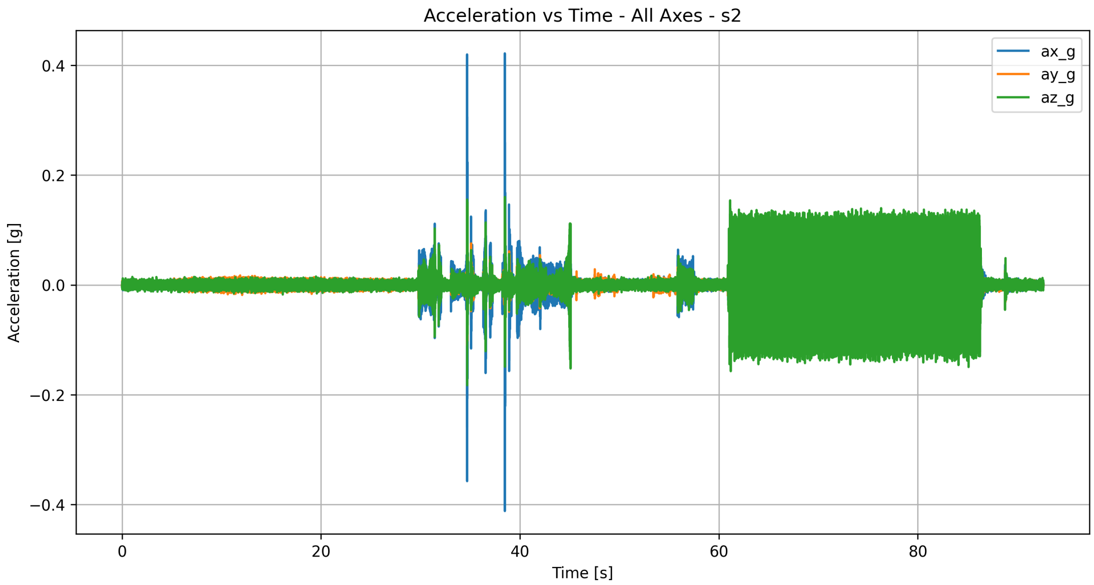

Figure 2: Acceleration vs time – all axes – Sensor 2.

Sensor 2 showed larger vibration amplitudes than Sensor 1. This suggests that the upper frame region is more flexible and experiences more vibration amplification than the base.

3.3 Sensor 3 – Extruder

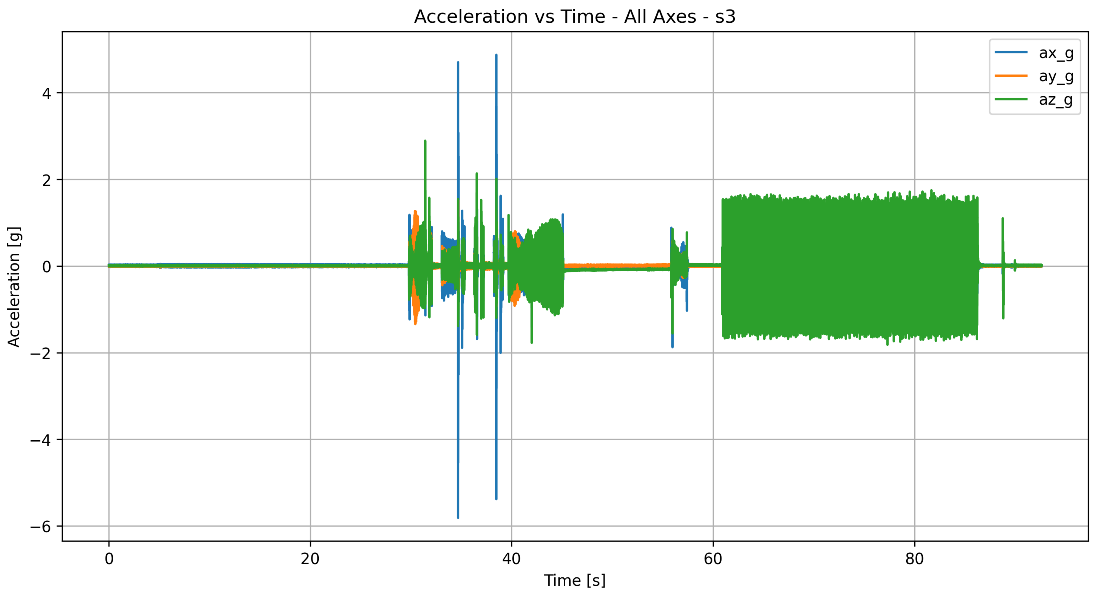

Figure 3: Acceleration vs time – all axes – Sensor 3.

Sensor 3 showed the largest vibration response. This is expected because the extruder is the moving component of the printer and is directly affected by motor motion, belt motion, inertia, and direction changes.

The largest measured peak acceleration was found in the X-axis of Sensor 3:

$$a_{peak} = 5.815946g$$

The peak-to-peak acceleration for this same axis was:

$$a_{p-p} = 10.692871g$$

This indicates that the extruder assembly is the most dynamically active part of the system.

4 FFT Method

The Fast Fourier Transform was used to convert the acceleration signal from the time domain into the frequency domain. The FFT allows the dominant vibration frequencies to be identified.

The general form of the discrete Fourier transform is:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N}$$

where:

- x_n is the acceleration signal in the time domain,
- X_k is the frequency-domain result,
- N is the number of samples,
- k is the frequency bin index.

5 FFT Results by Sensor

5.1 Sensor 1 FFT

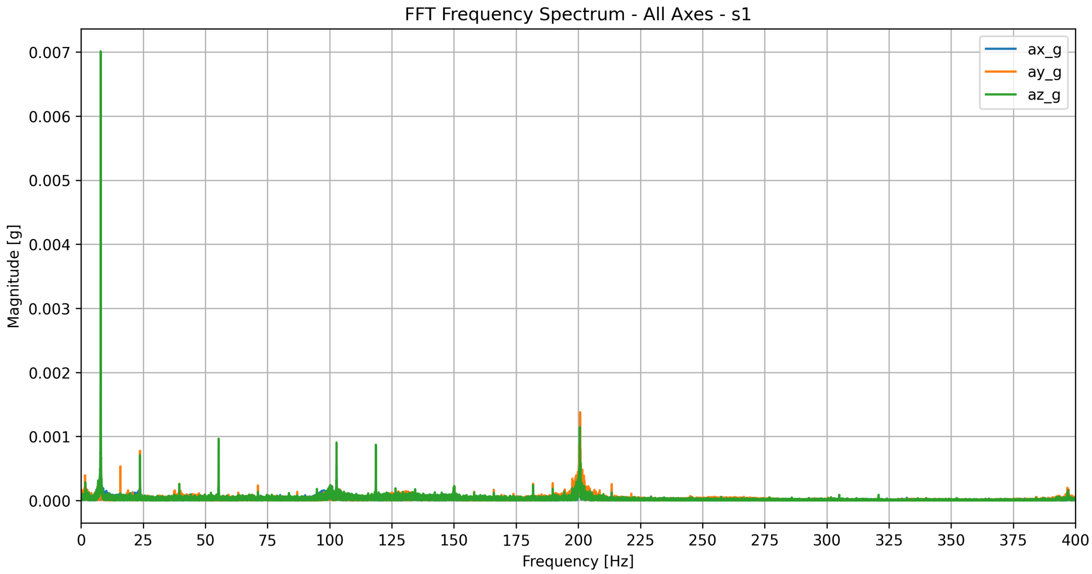

Figure 4: FFT frequency spectrum – all axes – Sensor 1.

Sensor 1 had its dominant frequency at approximately:

$$f = 7.90 \text{ Hz}$$

This frequency appeared in the X, Y, and Z axes. The strongest magnitude for Sensor 1 occurred in the Z-axis:

$$A = 0.007012g$$

Important secondary peaks were observed near 23.72 Hz, 55.34 Hz, 102.76 Hz, 118.58 Hz, and 200.63 Hz.

5.2 Sensor 2 FFT

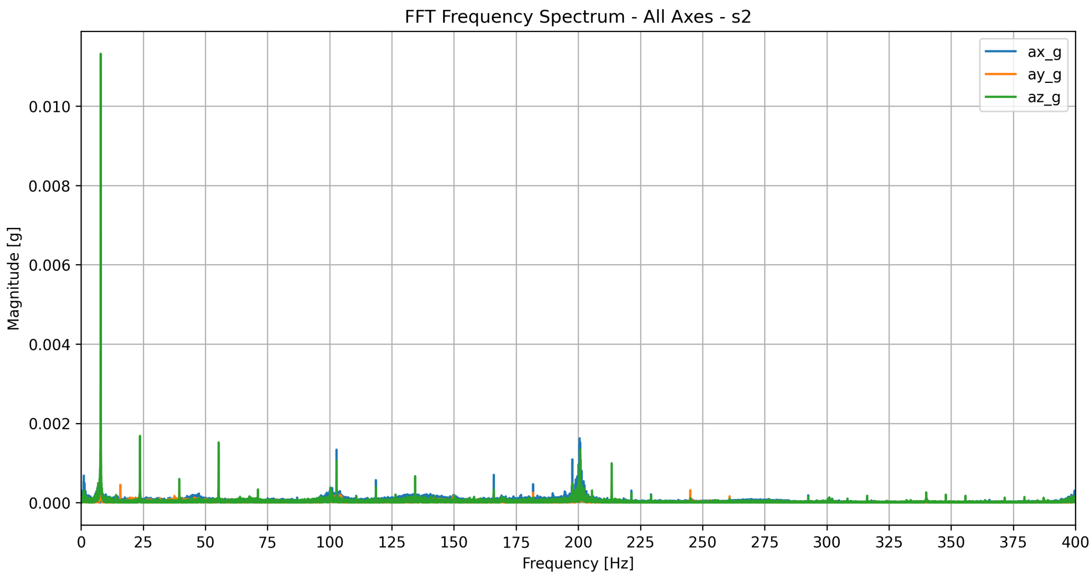

Figure 5: FFT frequency spectrum – all axes – Sensor 2.

Sensor 2 also showed a dominant frequency near:

$$f = 7.90 \text{ Hz}$$

The strongest Sensor 2 response occurred in the Z-axis:

$$A = 0.011320g$$

This shows that the upper frame has a stronger vibration response than the base, especially in the vertical direction.

5.3 Sensor 3 FFT

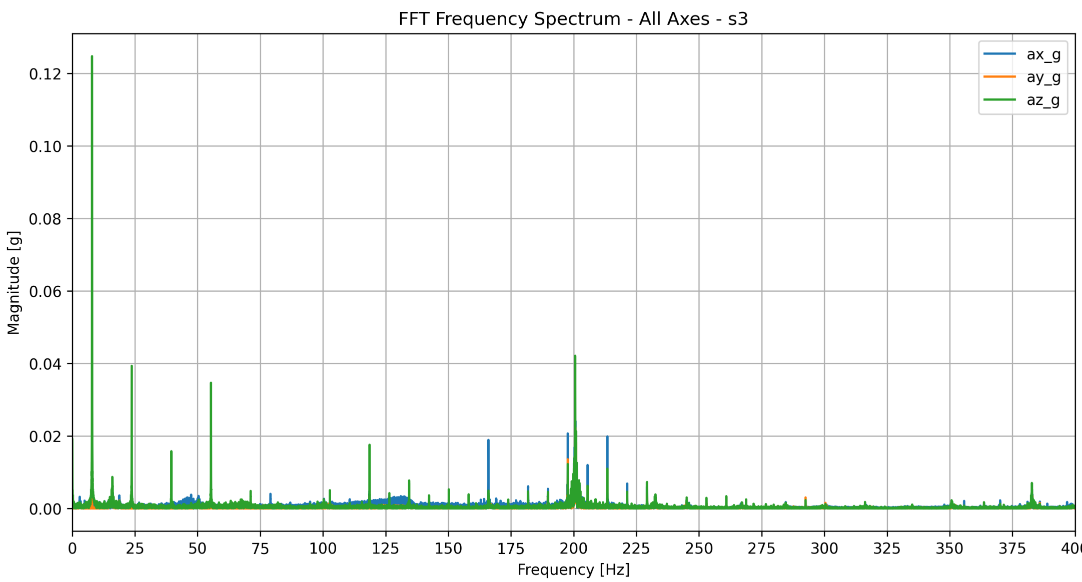

Figure 6: FFT frequency spectrum – all axes – Sensor 3.

Sensor 3 showed the largest FFT magnitudes. Two major frequency regions were observed:

- A low-frequency response near 7.90 Hz.
- A high-frequency response near 200 Hz.

The strongest overall FFT peak occurred in Sensor 3 Z-axis:

$$f = 7.90 \text{ Hz}$$

$$A = 0.124771g$$

Sensor 3 X and Y axes had their dominant frequencies near 200 Hz:

$$f_{s3,x} = 200.70 \text{ Hz}$$

$$f_{s3,y} = 200.64 \text{ Hz}$$

This suggests that the extruder assembly has strong high-frequency vibration components, likely related to motor excitation, belt dynamics, or local print-head vibration.

6 Overlapped FFT Comparison

6.1 X-Axis Overlap

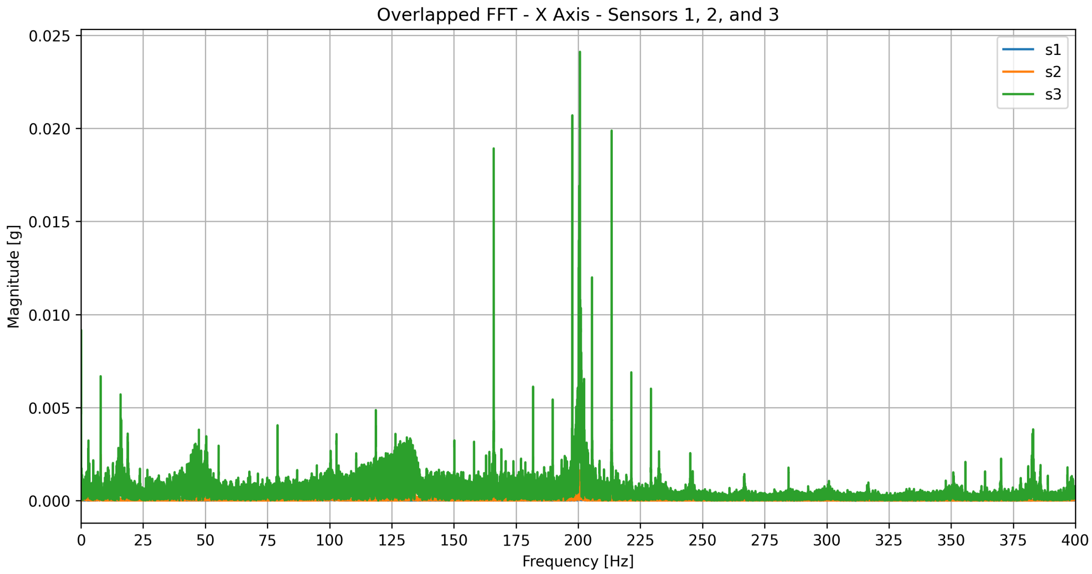

Figure 7: Overlapped FFT – X-axis – Sensors 1, 2, and 3.

The X-axis FFT comparison shows that Sensor 3 dominates the vibration response. The strongest peaks are concentrated around 200 Hz. Sensors 1 and 2 show much lower amplitudes, meaning that most of the high-frequency X-axis vibration is localized at the extruder.

6.2 Y-Axis Overlap

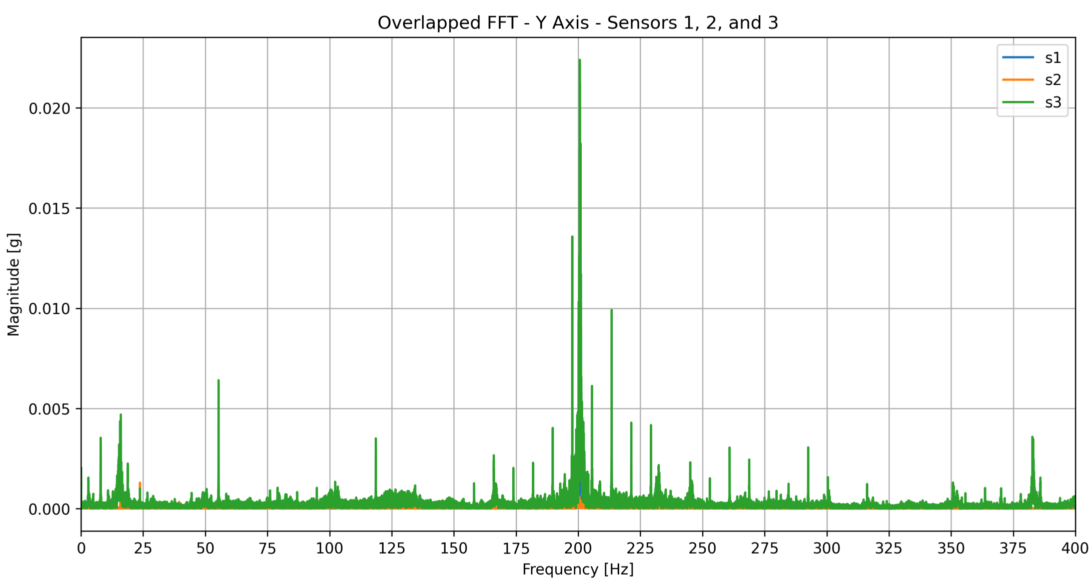

Figure 8: Overlapped FFT – Y-axis – Sensors 1, 2, and 3.

The Y-axis result also shows a strong Sensor 3 response near 200 Hz. The frame-mounted sensors remain much lower in amplitude. This supports the idea that the extruder experiences localized dynamic motion that is not fully transmitted to the frame.

6.3 Z-Axis Overlap

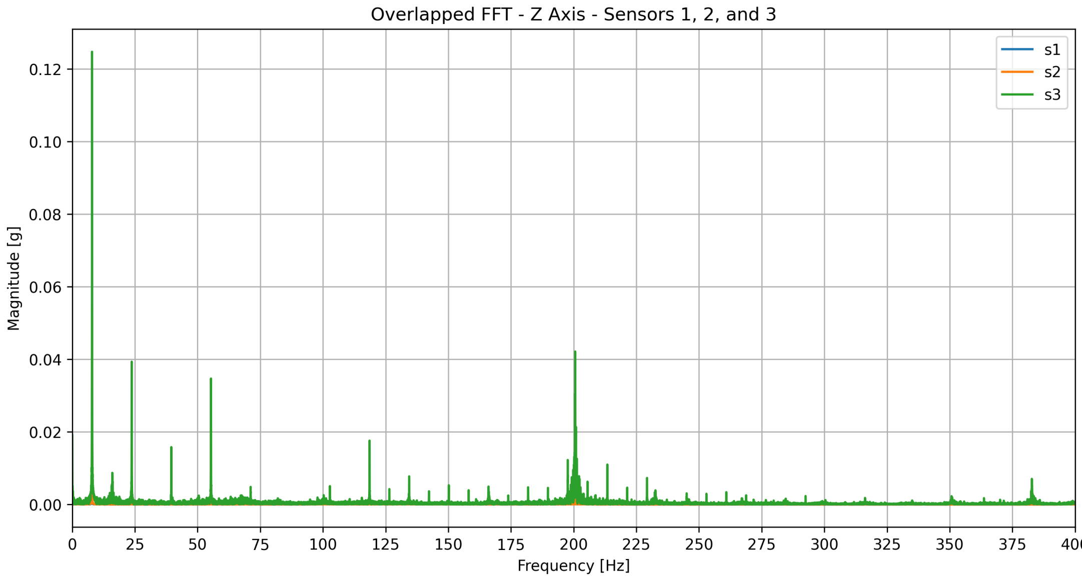

Figure 9: Overlapped FFT – Z-axis – Sensors 1, 2, and 3.

The Z-axis showed the largest overall vibration amplitude. The largest peak occurred near 7.90 Hz at Sensor 3. This suggests that the vertical direction is strongly affected by a low-frequency structural mode.

7 Interpretation of the 20 Hz Forced Vibration Test

Although the test was performed with a forced vibration input of approximately:

$$20 \text{ Hz}$$

the dominant frequency response did not occur exactly at 20 Hz. Instead, the strongest responses occurred near:

$$7.90 \text{ Hz}$$

and:

$$200 \text{ Hz}$$

This does not necessarily mean the test failed. A mechanical structure does not always respond most strongly at the applied forcing frequency. If the forcing excites a natural mode of the structure, the response may be larger at the natural frequency than at the input frequency.

The peak near 23.72 Hz is close to the intended 20 Hz forcing frequency and likely represents the direct forced response or a nearby excitation component. However, the much stronger 7.90 Hz and 200 Hz peaks suggest that the structure is amplifying other modes.

Possible explanations include:

- The printer has a low-frequency structural mode near 8 Hz.
- The extruder or gantry has a high-frequency local mode near 200 Hz.
- The 20 Hz input may be exciting harmonics or coupled structural modes.
- The printer motion system may introduce additional frequencies through motor steps, belts, or control behavior.

8 Summary of Main FFT Results

Sensor/Axis	Dominant Frequency [Hz]	Magnitude [g]	RMS [g]
s1 X	7.90	0.002297	0.010601
s1 Y	7.90	0.002676	0.014357
s1 Z	7.90	0.007012	0.027100
s2 X	7.90	0.002497	0.017143
s2 Y	7.90	0.003126	0.015194
s2 Z	7.90	0.011320	0.043078
s3 X	200.70	0.024116	0.211068
s3 Y	200.64	0.022395	0.140092
s3 Z	7.90	0.124771	0.533090

Table 2: Main FFT and RMS results by sensor and axis.

9 Acceleration Transmissibility

Acceleration transmissibility was calculated using Sensor 3 as the reference input because it was mounted on the extruder. Sensors 1 and 2 were treated as response sensors mounted on the frame.

$$T = \frac{A_{response}}{A_{reference}}$$

where:

- $A_{response}$ is the FFT magnitude measured at Sensor 1 or Sensor 2,
- $A_{reference}$ is the FFT magnitude measured at Sensor 3.

9.1 X-Axis Transmissibility

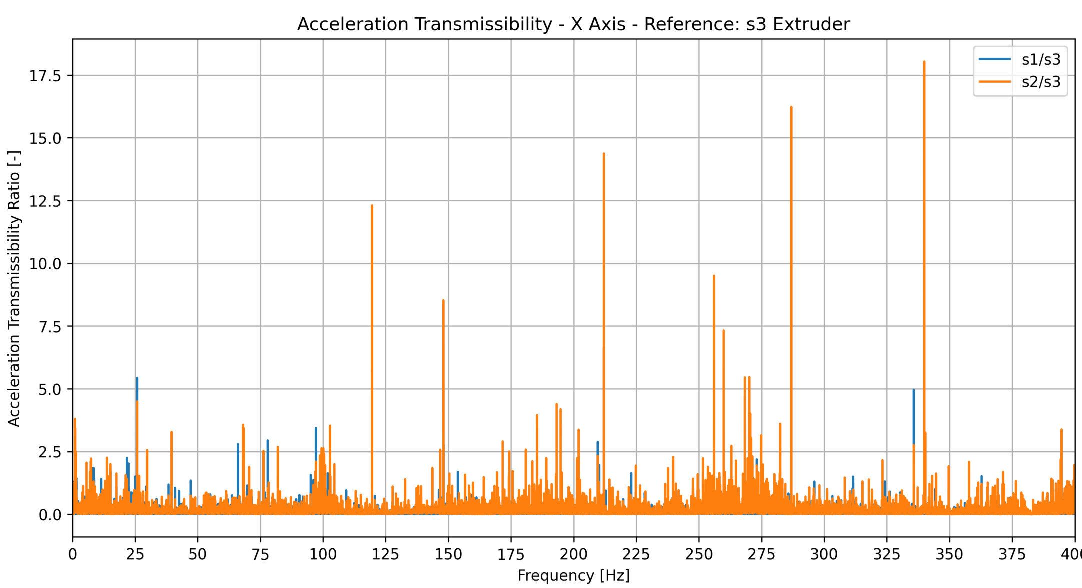

Figure 10: Acceleration transmissibility – X-axis – reference Sensor 3.

The X-axis transmissibility was generally below 1 for most frequencies. This means that most X-axis vibration measured at the extruder was attenuated before reaching the frame.

9.2 Y-Axis Transmissibility

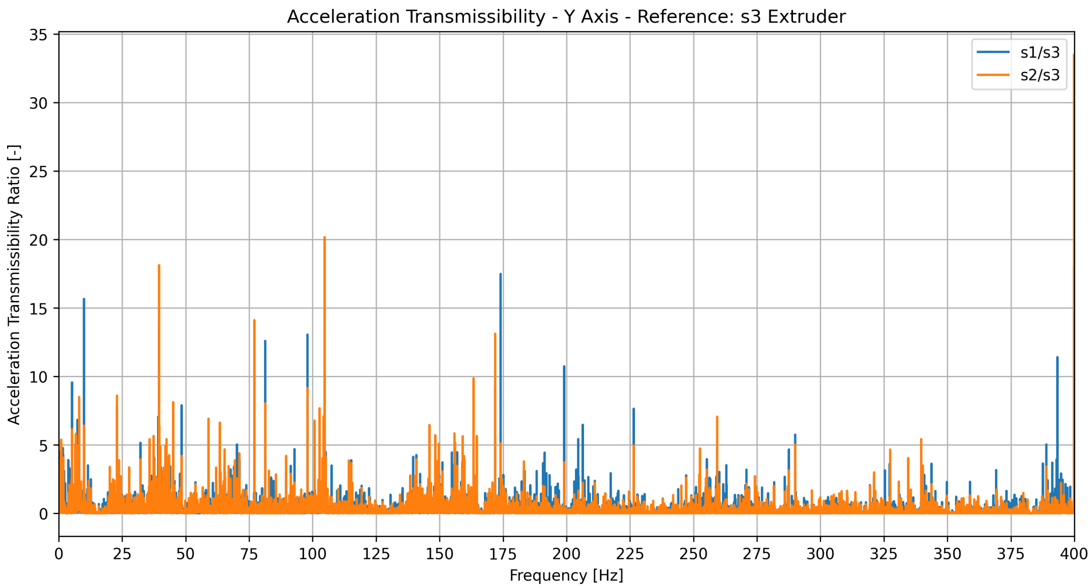

Figure 11: Acceleration transmissibility – Y-axis – reference Sensor 3.

The Y-axis transmissibility showed several sharp peaks. Some high transmissibility values occurred near the high-frequency end of the spectrum. These peaks may represent localized frame amplification or may occur when the reference signal from Sensor 3 is very small at certain frequencies.

9.3 Z-Axis Transmissibility

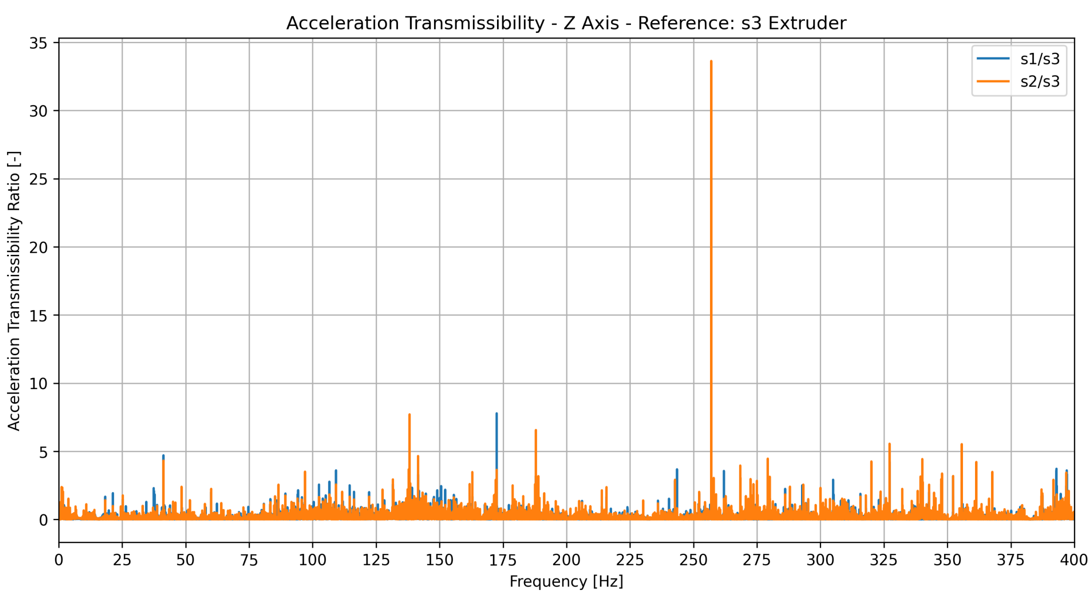

Figure 12: Acceleration transmissibility – Z-axis – reference Sensor 3.

The Z-axis transmissibility also showed sharp localized peaks. However, for most of the frequency range, the frame sensors remained lower than the extruder response. This indicates that the strongest vertical vibration remains concentrated at the extruder.

10 Transmissibility Summary

Ratio	Maximum T	Frequency [Hz]	Mean T
s1/s3 X	6.6584	212.09	0.0680
s2/s3 X	18.0361	339.94	0.1296
s1/s3 Y	18.0409	399.86	0.2134
s2/s3 Y	33.5188	399.86	0.1882
s1/s3 Z	12.2423	256.99	0.1060
s2/s3 Z	33.6335	256.99	0.1292

Table 3: Acceleration transmissibility summary.

The average transmissibility values are all below 1. This means the frame generally receives less vibration than the extruder. However, the maximum transmissibility values are much larger than 1 at specific frequencies. These isolated peaks require further study because they may correspond to structural resonances or numerical effects caused by low reference amplitude.

11 Key Observations

1. The ESP32-based system successfully collected data at approximately 800 Hz.
2. The sampling quality was stable, with only one estimated dropped sample.
3. The dominant low-frequency structural response appeared near 7.90 Hz.
4. The extruder showed strong high-frequency vibration near 200 Hz.
5. Sensor 3 had much larger vibration amplitudes than Sensors 1 and 2.
6. The Z-axis produced the strongest low-frequency response.
7. The forced 20 Hz excitation was visible near 23.72 Hz, but it was not the dominant response.
8. The frame attenuated most of the vibration coming from the extruder, except at isolated frequencies.

12 Conclusions

The updated vibration test shows that the low-cost ESP32 and MPU6050-based system is capable of identifying meaningful vibration behavior in a 3D printer structure.

The main finding is that the printer does not respond most strongly at the applied 20 Hz forcing frequency. Instead, the structure appears to amplify vibration near 7.90 Hz and near 200 Hz.

The 7.90 Hz response likely represents a low-frequency structural mode of the printer. The 200 Hz response appears to be strongly localized at the extruder and may be related to the motion system, motor excitation, belt behavior, or local print-head vibration.

The transmissibility analysis shows that most vibration is attenuated as it travels from the extruder to the frame. However, sharp transmissibility peaks suggest that there are specific frequencies where the frame may amplify vibration.

Overall, the results support the continued development of this project as a low-cost structural vibration analysis platform for educational and experimental use.

13 Recommended Next Steps

- Repeat the test at different forced frequencies such as 2 Hz, 10 Hz, 50 Hz, 100 Hz, 150 Hz, and 200 Hz.
- Perform a controlled frequency sweep to better identify natural frequencies.
- Compare the MPU6050 results with a higher-quality accelerometer in the future.

- Add a band-limited transmissibility analysis to avoid misleading peaks caused by small denominator values.
- Investigate the strong 200 Hz response at the extruder.
- Improve the report plots by adding annotations for the 7.90 Hz, 20 Hz, 23.72 Hz, and 200 Hz regions.

Source File

The numerical results used in this report were taken from:

`summary_V2.txt`